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Zachary Bednarke

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EDUCATION

UNIVERSITY OF WASHINGTON

MSc - Physics | 2017 - 2019
PhD Physics | 2017 -> Kernel
Thesis: Deep-brain source reconstruction in magnetoencephalography

Significant Coursework
Statistical Physics & Deep Learning
CSE 546: Machine Learning
Learning in the Immune System
Machine Learning in Physics

VANDERBILT UNIVERSITY

Music, Physics | 2013 - 2017
GPA:3.93 | Honors in Physics
Thesis: On Relativistic Beltrami Fluids
Dean's Honor Scholarship
Academic Achievement Scholarship

ARROWHEAD HIGH SCHOOL

GPA:4.0 | SAT 2340
Winner: UW Full Tuition Music Scholarship Competition
National Merit Scholarship | >99.5 pctile

Links
First Phys/Mus major @ VU
Kernel Flux
Linkedin

Patent Links
MEG Calibration
Physics Based Data Cleansing
Estimating Brain Signal at a Distance
+2 more pending publication (2021)

Academic Publications
1 on Kernel Flux
2 on high-energy physics
1 on information theory in MEG

PROJECTS

Soliton Genesis
Automated trading advisor. SG focuses on short-volatility opportunities in options markets.

Fitness Forecast
Based on a long term plan, short term sensor data is analyzed daily to distinguish material plan deviations from negligible anomalies

EXPERIENCE

HEDGE FUND | GM Technologies | Quantitative Researcher

November 2021 - Present | Los Angeles, CA

- Employed on contract as a software developer / quantitative researcher
- Developed a modeling & backtesting suite in Python from the ground up
- Developed real-time data & signal visualization tools (PyQT and PyQTGraph)
- Improved GMT's live-trading software system. (Python socket server interfacing with Interactive Brokers REST API)
- Database design and engineering (PostgreSQL)
- My contributions to the composite algorithm yielded improvements in real 2022 trading relative to hypothetical returns of the prior model

KERNEL | Neural Device Engineer | Data Scientist

March 2019 - November 2021 | Los Angeles, CA

- Team Lead: Magnetic Image Processing Team
- Developed analysis pipelines in Python with a 6 person team of data scientists, neuroscientists, and physicists
- Developed and maintained (PyTorch & MLFlow) deep neural networks that cleaned Flux data, inferred human brain states, and localized brain activity
- Processed over 1000 hours of human brain data recorded by Flux
- Made major contributions to the design of the **Kernel Flux** system. I produced 5 patent applications
- Developed computer vision programs with OpenCV for subject monitoring & pose estimation during neural data collection
- Produced control software that interfaced with Flux's FPGA controller
- The first Kernel Flux system shipped in Q4 2021

I-LABS MEG Physics Group | PhD Thesis Work

2017 - 2020 | Seattle, WA

- Studied magnetoencephalography MEG under **Samu Taulu** at the **UW Institute for Learning and Brain Sciences**.
- Inferred brain-region connectivity networks related to the mechanisms of learning in the brains of infants
- Developed novel machine learning algorithms for MEG connectivity analysis
- Invented novel methods for reconstructing 3D distributions of brain activity from sparse sampling grids
- Designed next-gen magnetometer arrays with sampling geometries optimized for these new source reconstruction methods

VU Particle and Field Theory Group | Thesis Research

October 2015 - April 2017 | Nashville, TN

- I worked with **Thomas Kephart** and **Roman Buniy** to develop the Beltrami equation for fluids in Minkowski spacetime
- I presented a proof of the existence of this class of fluid flows at the 2018 Shanks Workshop on mathematical fluid dynamics

VU High Energy Physics Group

April 2015 – August 2015 | Nashville, TN

- With **Alfredo Gurrola**, **VU JSWHEP** and the CMS analysis group, I developed tools to aid in analysis of high energy physics data
- Our analysis work yielded two publications